WASP-12b

R-1834

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-12_160131` · created 2026-08-03T06:11:23+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457418.70215 +/- 0.00025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.125 +/- 0.029
Transit depth (Rp/Rs)^2	1.56 +/- 0.72 [%]
Orbital Inclination (inc)	86.46 +/- 2.66
Airmass coefficient 1 (a1)	1.086 +/- 0.019
Airmass coefficient 2 (a2)	-0.077 +/- 0.05
Scatter in the residuals of the lightcurve fit is	1.77 %
Best Comparison Star	#1 - [491, 393]
Optimal Aperture	3.55
Optimal Annulus	7.4
Transit Duration (day)	0.1289 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.125 ± 0.029 vs 0.1178 (deviation 0.25σ)
Duration fit vs published	0.1289 d vs 0.125 d (deviation 0.8σ)
Mid-time O–C	-0.6 min
Depth SNR	4.3
Residual scatter	1.77 %

Light curve

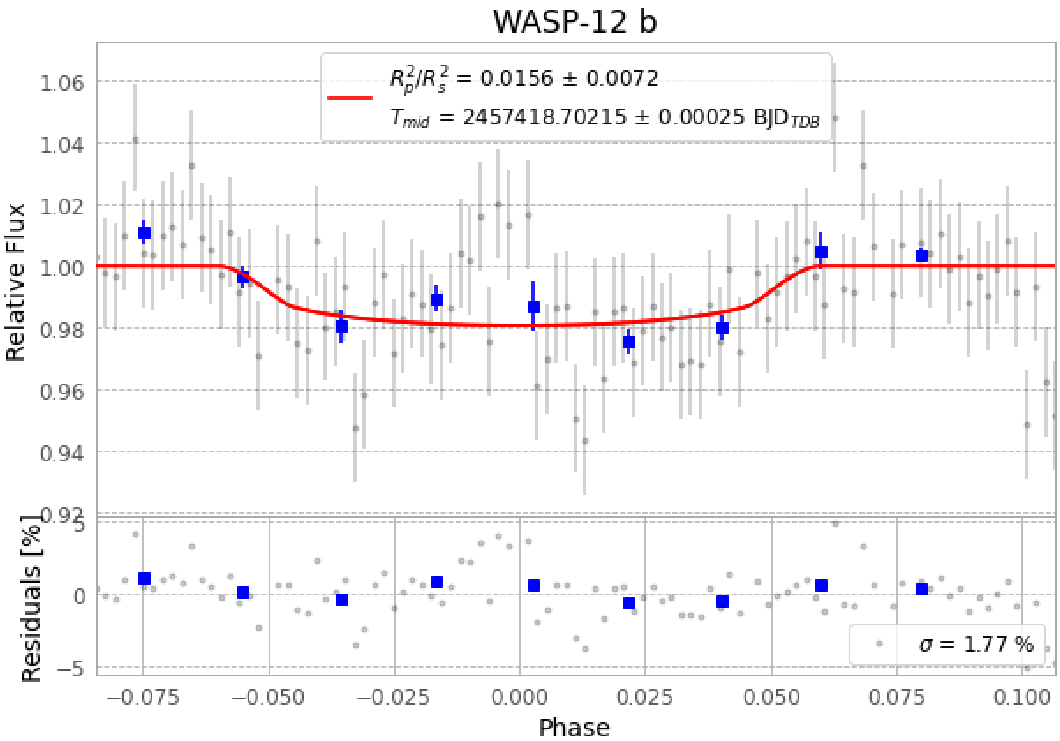

Figure 1 — FinalLightCurve_WASP-12 b_2016-01-30.png (SHA-256 653e79847521cff0…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [260, 213], 1 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 c4b0271582717922…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

101 FITS frames (66 MB), `WASP-12160131023012.FITS` → `WASP-12160131073012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-160131.log (915edb37b0e5…), FinalLightCurve_WASP-12 b_2016-01-30.png (653e79847521…), FinalLightCurve_WASP-12 b_2016-01-30.csv (89a12b1ecab6…)

Compute resources

Wall time 145.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)